MINISTRY OF AGRICULTURE AND RURAL DEVELOPMENT
VIETNAM NATIONAL UNIVERSITY OF AGRICULTURE

SOCIALIST REPUBLIC OF VIETNAM
Independent –Freedom – Happiness

PROGRAM FOR UNDERGRADUATE
EDUCATION PROGRAM: VETERINARY MEDICINE
MAJOR 1: VETERINARY MEDICINE
MAJOR 2: ANIMAL HUSBANDRY

COURSE SPECIFICATION
TY03053: ANIMAL REPRODUCTION 1
(SINH SẢN GIA SÚC 1)

I. General information
- Term: 6
- Course credits: Total credits 02 (Lecture: 1.5 - Practice: 0.5; Self-study: 6)
- Credits hours for teaching and learning activities:
  + In-class lecture: 22
  + Lab practical: 8
- Self-study: 90 hours *(following personal's plan or lecturer's instruction)*
- Department conducting the course:
  - Department: Surgery-Reproduction
  - Faculty: Veterinary Medicine
- Kind of course:

Foundation □		Fundamental □		Major 1 ☒		Major … □	
Compulsory □	Elective □	Compulsory □	Elective □	Compulsory ☒	Elective □	Compulsory □	Elective □

- Parallel learning course:
- Prerequisite learning course: Animal physiology 1 (CN02303)
- Course language: English ☐ Vietnamese ☒

II. Output standards of the training program
**** Output standards and indicators of the training program to which the course contributes:***

Output standards of the training program After completing the program, students can:	**Indicator of the output standard of the training program**
Expertise	
PLO3. Apply veterinary knowledge to the diagnosis and treatment of livestock diseases effectively.	3.2: Livestock treatment 3.3: Evaluating the effectiveness of diagnosis and treatment 3.4: Improve animal health

Output standards of the training program After completing the program, students can:	**Indicator of the output standard of the training program**
PLO4. Design programs for diagnosis and treatment of diseases for livestock, procedures for disease prevention for animals according to prescribed standards.	4.1: Design diagnostic and treatment programs for livestock diseases
General skills	
PLO5. Apply critical and creative thinking, evidence-based reasoning to solve problems effectively.	5.3: Demonstrate problem-solving abilities including identifying problems, knowing when and how to gather information, and evaluating and selecting information needed for problem solving
Advanced skill	
PLO10. Using information technology and modern equipment of the veterinary industry for the diagnosis, prevention and treatment of animal diseases and management of animal diseases to achieve the set goals.	10.2: Using modern veterinary equipment for diagnosis, treatment and management of animal diseases to achieve the set goals
Autonomy and responsibility	
PLO12. Comply with regulations and laws, maintain professional ethics.	12.2: Maintain the professional ethics of veterinarians
PLO13. Take responsibility to protect the environment, improve human health and show love for animals.	13.2: Show love for animals

III. Course objectives and Expected learning outcomes

**** Course objectives***:

The course aims to provide students with knowledge and skills on animal reproduction and reproductive techniques applied on cattle such as intrauterine insemination techniques, techniques for the use of reproductive hormones in animal husbandry – veterinary, embryo transmission technology and sex control in animal reproduction.

**** Course expected learning outcomes:***

The study contributes to the following output standard of the training program according to the following level:

I – Introduction (Giới thiệu); P – Practice (Thực hiện); R – Reinforce (Củng cố); M – Master (Đạt được)

Code	Course	The level of contribution of the module to the output standard of the training program										
		3.2	3.3	3.4	4.1	5.3	10.2	12.2	13.2	…	…	…
TY03053	Animal reproduction 1	R	P	R	P	R	R	R	P			

	Course expected learning outcomes After successfully completing this course you should be able to	**Output standard indicator of the training program**

Knowledge		
K1	Applying factors affecting the diagnosis and treatment of reproductive diseases for livestock, strategies for prevention and treatment of reproductive diseases for animals and public health	3.2. Evaluation of the effectiveness of diagnosis and treatment of diseases in livestock
K2	Predictive assessment of the effectiveness of diagnosis and treatment of reproductive diseases in livestock, strategies for prevention of reproductive diseases in animals and public health	3.3. Evaluation of the effectiveness of diagnosis and treatment of diseases in livestock
K3	Apply specialized knowledge to develop diagnostic and treatment programs for livestock diseases, strategies for prevention of reproductive diseases for animals and public health to ensure economic and social benefits, environment and animal welfare.	3.4. Evaluation of the effectiveness of diagnosis and treatment of diseases in livestock
Skill		
K4	Test clinical and non-clinical skills, common technical procedures in the diagnosis and treatment of reproductive diseases for livestock, prevention of reproductive diseases for animals to contribute to the protection of public health	5.3. Apply critical and creative thinking to solve problems effectively
K5	Coordinating commonly applied techniques and technologies in reproduction to improve reproductive productivity	10.2. Using information technology and modern equipment of the veterinary industry for the diagnosis, prevention and treatment of animal diseases and management of animal diseases to achieve the set goals.
Autonomy and responsibility		
K6	Endorsing behaviors that comply with laws, internal regulations and professional ethical standards	12.2. Practice compliance with regulations and laws, maintain professional ethics
K7	Connecting behaviors that protect the environment, protect public health and respect animal welfare	13.2. Practice responsibility to protect the environment, improve human health and show love for animals

IV. Course description

Code: TY03053. Name: Animal reproduction 1 (Sinh sản gia súc 1). (Total credits 02: lecture 1,5 – practice 0,5; self-study 6).

Content: The module describes the biological nature of sexual reproduction, the physiological mechanisms regulating sexual reproduction, and maturation in mammals. Reproductive activity of male and female cattle, physiology of fertilization, reproductive hormones and applications in animal husbandry - veterinary medicine. Techniques for intrauterine insemination of cattle; embryo transfer technology and an overview of sex control in animal reproduction.

Teaching methods: Students listen to lectures and practice in class in combination with self-study, self-referencing documents and exchanging with friends and teachers. Students practice under the guidance of teachers.

Assessment methods: Attendance: 10%, Mid-term exam: 30%, Final exam: 60%. Prerequisite learning course: Animal physiology 1 (Sinh lý động vật 1 - CN02303)

V. Teaching and learning methods

1. Teaching methods

Table 1: Teaching methods

Course expected learning outcomes / **Teaching methods**	**K1**	**K2**	**K3**	**K4**	**K5**	**K6**	**K7**
Presentations	x	x	x	x	x	x	x
Practice				x	x	x	x

2. Learning methods

Students follow the lecturer's presentation and interact with the lecturer.

VI. Tasks of students

- Attendance: Students must attend at least 75% of the theoretical duration to be eligible for the final exam.
- Practice: All students attending this section must have sufficient number of practice lessons.
- Mid-term exam: Students must take the mid-term exam.
- Final exam: Students must take the final exam.

VII. Assessment methods

1. Grading: 10

2. The average score of the course is the sum of the rubrics' scores multiplied by the respective weights of each rubric

3. Weighting:

- Attendance: 10 %
- Process score/Mid-term exam: 30%
- Final exam: 60%

Table 2. Rating Matrix

Course expected learning outcomes	K1	K2	K3	K4	K5	K6	**K7**	**Time/week of study**
Process assessment (50%)								
Rubric 1. Attendance (10%)	x	x	x	x	x	x	x	Week 1-5

Rubric 2. Practice (no weighting)				x	x	x	x	Week 3-11
Rubric 3. Mid-term exam (30%)	x	x						Week 4
Final exam (50%)								
Rubric 4. Final exam (60%)	x	x						According to the academy's exam schedule

Rubric 1: Class attendance_10%)

Criteria	Weighting (%)	Excellence 100%	Good 75%	Fair 50%	Poor 0%
Level of engagement and behaviour	50	Always listening attentively and contributing actively to class's activities	Mostly listening attentive and contributing to class's activities	Listening attentively	Not listening attentively
Participant	50				

Rubric 2: Practice and report (0 %, no weighting, no points.)

Criteria	Weighting (%)	Excellence 100%	Good 75%	Fair 50%	Poor 0%
Level of engagement and behaviour	10	Actively provoke problems, questions and share ideas	Contribute to discussion and share ideas	Sometimes contribute to discussions and share ideas	No contribution to discussion and share ideas
Practical results	60	Practical results are complete and meet all requirements	Practical results are complete and meet most requirements with a minor mistake	Practical results are complete and meet most requirements but have a serious mistake	Practical results are incomplete and do not meet any requirements
Practical report	30	Format and styles are as required; report is submitted on time	Marks are given according to level of meeting the requirements		

Rubric 3: Mid-term exam

Topics/Contents	Indicators	Course expected learning outcomes
Concepts of reproduction in cattle	Indicator 1: The concept of cattle maturation Indicator 2: Regulatory mechanisms of reproductive activity in cattle Indicator 3: Estrous cycle in female cattle Indicator 4: Sperm morphology, structure and characteristics Indicator 5: Semen Indicator 6: Seminal plasma Indicator 7: Physiology of fertilization Indicator 8: Method of insemination Indicator 9: Movement of eggs, sperm and sperm survival time in female genitalia Indicator 10: Fertilization process	K1, K2, K3, K6, K7
Reproductive techniques in cattle	Indicator 11: Concept of intrauterine insemination, advantages and disadvantages of cattle intrauterine insemination Indicator 12: Methods of extraction of cattle semen Indicator 13: Criteria for testing cattle semen Indicator 14: Methods of preparation and preservation of cattle semen Indicator 15: Cattle insemination method Indicator 16: Concept of cattle embryo transfer technique Indicator 17: Scientific basis of cattle embryo transfer technique Indicator 18: Main contents of cattle embryo transfer technique Indicator 19: Embryo transfer technology in cows Indicator 20: Technology of embryo transfer in goats - sheep Indicator 21: Embryo transfer technology in buffalo Indicator 22: Embryo transfer technology in horses Indicator 23: Technology of embryo transfer in pigs	K1, K2, K3, K4, K5, K6, K7
Application of hormones in animal reproduction	Indicator 24: Certain major hormones and their preparations for reproductive applications Indicator 25: Some hormone applications to improve reproductive productivity	K1, K2, K3, K6, K7

Rubric 4: Final exam (60%)

Topics/Contents	Indicators	Course expected learning outcomes
Concepts of reproduction in cattle	Indicator 1: The concept of cattle maturation Indicator 2: Regulatory mechanisms of reproductive activity in cattle Indicator 3: Estrous cycle in female cattle	K1, K2, K3, K6, K7

	Indicator 4: Sperm morphology, structure and characteristics Indicator 5: Semen Indicator 6: Seminal plasma Indicator 7: Physiology of fertilization Indicator 8: Method of insemination Indicator 9: Movement of eggs, sperm and sperm survival time in female genitalia Indicator 10: Fertilization process	
Reproductive techniques in cattle	Indicator 11: Concept of intrauterine insemination, advantages and disadvantages of cattle intrauterine insemination Indicator 12: Methods of extraction of cattle semen Indicator 13: Criteria for testing cattle semen Indicator 14: Methods of preparation and preservation of cattle semen Indicator 15: Cattle insemination method Indicator 16: Concept of cattle embryo transfer technique Indicator 17: Scientific basis of cattle embryo transfer technique Indicator 18: Main contents of cattle embryo transfer technique Indicator 19: Embryo transfer technology in cows Indicator 20: Technology of embryo transfer in goats - sheep Indicator 21: Embryo transfer technology in buffalo Indicator 22: Embryo transfer technology in horses Indicator 23: Technology of embryo transfer in pigs	K1, K2, K3, K4, K5, K6, K7
Application of hormones in animal reproduction	Indicator 24: Certain major hormones and their preparations for reproductive applications Indicator 25: Some hormone applications to improve reproductive productivity	K1, K2, K3. K6. K7

Table 3. Indicators of course expected learning outcomes

Course expected learning outcomes	**Indicators of expected learning outcomes**
K1. Applying factors affecting the diagnosis and treatment of reproductive diseases for livestock, strategies for prevention and treatment of reproductive diseases for animals and public health	Indicator 1: Estrous cycle in female cattle Indicator 2: Sperm morphology, structure and characteristics Indicator 3: Semen Indicator 4: Seminal plasma Indicator 5: Physiology of fertilization Indicator 6: Concept of intrauterine insemination, advantages and disadvantages of cattle intrauterine insemination Indicator 7: Certain major hormones and their preparations for reproductive applications Indicator 8: Some hormone applications to improve reproductive productivity

K2. Predictive assessment of the effectiveness of diagnosis and treatment of reproductive diseases in livestock, strategies for prevention of reproductive diseases in animals and public health	Indicator 9: Method of insemination Indicator 10: Movement of eggs, sperm and sperm survival time in female genitalia Indicator 11: Fertilization process
K3. Apply specialized knowledge to develop diagnostic and treatment programs for livestock diseases, strategies for prevention of reproductive diseases for animals and public health to ensure economic and social benefits. , environment and animal welfare.	Indicator 12: The concept of cattle maturation Indicator 13: Regulatory mechanisms of reproductive activity in cattle
K4. Test clinical and non-clinical skills, common technical procedures in the diagnosis and treatment of reproductive diseases for livestock, prevention of reproductive diseases for animals to contribute to the protection of public health	Indicator 14: Methods of extraction of cattle semen Indicator 15: Criteria for testing cattle semen Indicator 16: Methods of preparation and preservation of cattle semen Indicator 17: Cattle insemination method Indicator 18: Concept of cattle embryo transfer technique Indicator 19: Scientific basis of cattle embryo transfer technique Indicator 20: Main contents of cattle embryo transfer technique
K5. Coordinating commonly applied techniques and technologies in reproduction to improve reproductive productivity	Indicator 21: Embryo transfer technology in cows Indicator 22: Technology of embryo transfer in goats - sheep Indicator 23: Embryo transfer technology in buffalo Indicator 24: Embryo transfer technology in horses Indicator 25: Technology of embryo transfer in pigs
K6. Endorsing behaviors that comply with laws, internal regulations and professional ethical standards	Indicator 26: Practice practical skills on animals in accordance with current veterinary laws
K7. Connecting behaviors that protect the environment, protect public health and respect animal welfare	Indicator 27: Practice practical skills with gentleness, care, guaranteed to cause the least amount of pain to the animal

4. Course requirements and policies:

Late submission of assignments: All late submissions will result in a deduction of 10% of the score.

Taking exams: Students must take the midterm exam. Students who do not take the midterm exam will not be able to take the final exam.

Participating in practice: If students do not participate in enough practice content, they will not be able to participate in the final exam.

Ethical requirements: Students must abide by the rules of the subject.

VIII. Text books and references

**** Text Books/Lecture Notes:***

- Trần Tiến Dũng Dương Đình Long, Nguyễn Văn Thanh (2002). Sinh sản gia súc. NXB Đại học Nông nghiệp, Hà Nội. 230 trang.

- Nguyễn Văn Thanh (chủ biên). 2016. Công nghệ sinh sản vật nuôi. NXB Đại học Nông nghiệp. Hà Nội. 346 trang

**** Additional references:***

- Geaffrey H. Antur Bailliere Tindall. 1975. Veterinary reproduction and obstetrics.
- Ian Gordon. 1996. Controlled reproduction in farm animals. Dublin - Ireland - Lab. International.
- G.M. Stone and G. Evans. 1996. Animal reproduction: Research and practice. Elesevier – Netherland.

IX. Course outline

Week	Content	Course expected learning outcome
1	***Chapter 1: Animal maturity***	
	A/ Main contents: *(2 lesson)* **Theory:** 1.1. The biological nature of sexual reproduction 1.2. Physiological mechanisms regulating sexual reproduction 1.3. Maturity in mammals 1.4. Age of physical maturity of cattle **Practice/Experiment:** *(2 lesson)* - Indicators of maturation in male and female cattle	K1-K7
	B/ **Self- study contents:** *(3 lesson)* - Physiological mechanisms regulating sexual reproduction in cattle - Age of sexual maturity in cattle and factors affecting maturity	K1-K5
2	***Chapter 2: Reproductive activity of male cattle***	
	A/ Main contents: *(2 lesson)* **Theory:** 2.1. Anatomy of the male cattle genitalia 2.2. Semen 2.3. Male reproductive cell (sperm) 2.4. Seminal plasma **Practice/Experiment:** *(2 lesson)* Anatomy of male cattle genitalia	K1-K7
	B/ **Self- study contents:** *(4 lesson)* - Anatomy of the male reproductive organs of some basic cattle species - Structure of male reproductive cells (sperm)	K1-K7
3	***Chapter3: Reproductive activity of female cattle***	
	A/ Main contents: *(2 lesson)* **Theory:** 3.1. Anatomy of the female cattle genitalia 3.2. Female reproductive cell 3.3. The reproductive cycle in female cattle	K1-K7

Week	Content	Course expected learning outcome
	Practice/Experiment: *(2 lesson)* Anatomy of female cattle genitalia	
	***B/* Self- study contents:** *(4 lesson)* - Anatomy of the female reproductive organs of some basic cattle species - Structure of female reproductive cells (ovary) - The reproductive cycle of female cattle: concepts, stages, characteristics of the reproductive cycle of some cattle species	K1-K7
4	***Chapter 4: Fertilization***	
	A/ Main contents: *(2 lesson)* **Theory:** 4.1. Characteristics and forms of mating 4.2. Method of insemination 4.3. The movement of eggs, sperm and the life time of sperm in the female genitalia 4.4. Fertilization **Practice/Experiment:** *(2 lesson)* - Types of mating in cattle	K1-K7
	***B/* Self- study contents:** *(4 lesson)* 4.5. Movement of eggs and sperm in the genital tract of female cattle; Sperm life time in female reproductive organs 4.6. Physiology of fertilization: concept, stages, meaning	K1-K7
5	***Chapter 5: Hormones and applications in reproduction***	
	A/ Main contents: *(2 lesson)* **Theory:** 5.1. Common problems 5.2. Some major hormones and their preparations for reproductive applications 5.3. Some hormone applications to improve reproductive productivity 5.4. Conclusion **Practice/Experiment:** *(2 lesson)* - Techniques for using hormones in animal reproduction	K1-K7
	***B/* Self- study contents:** *(4 lesson)* 5.5. Reproductive hormones: concepts, hormones involved in livestock reproduction. - Application of hormone preparations to improve productivity in livestock production	K1-K7

Week	Content	Course expected learning outcome
6	***Chapter 6: Cattle intrauterine insemination technique***	
	A/ Main contents: *(6 lesson)* **Theory:** 6.1. Common problems 6.3. Scientific basis of cattle intrauterine insemination technique 6.4. Economic and technical benefits of cattle intrauterine insemination 6.5. Techniques for extraction of cattle semen 6.6. Several factors mainly affect the quantity and quality of semen 6.7. Animal semen test 6.8. Preparation, preservation, and transportation of semen 6.9. Preservation of semen 6.10. Transportation of semen 6.11. Ejaculation **Practice/Experiment:** *(2 lesson)* - Method of extracting semen in cattle - Check cattle semen: color, smell, pH, vitality, concentration, percentage of morphological sperm, resistance. - Techniques for preserving cattle semen - Cattle insemination technique	K1-K7
	***B/* Self- study contents:** *(4 lesson)* - Semen extraction techniques in male cattle - Techniques to test cattle semen - Techniques for making media and how to prepare semen - Methods of preserving cattle semen - Method of insemination for some cattle species	K1-K7
7	***Chapter 7: Embryo transfer technology***	
	A/ Main contents: *(4 lesson)* **Theory:** 7.1. The development history of embryo transfer technology 7.2. Technology of bovine embryo transfer 7.3. Technology of embryo transfer in goats - sheep 7.4. Technology of embryo transfer in buffalo 7.5. Technology of embryo transfer in horses 7.6. Technology of embryo transfer in pigs **Practice/Experiment:** *(2 lesson)* - Method of inducing hyperovulation in cattle for embryos	K1-K7

Week	Content	Course expected learning outcome
	- Methods to induce estrus in synchrony in cattle receiving embryos - Method of collecting embryos from the uterus of cattle for embryos by non-surgical method through washing - Method of embryo transfer	
	***B/* Self- study contents:** *(4 lesson)* - Contents of bovine embryo transfer technology - Contents of embryo transfer technology in goats - sheep - Contents of embryo transfer technology in buffalo - The contents of embryo transfer technology in horses - The contents of embryo transfer technology in pigs	K1-K7
8	***Chapter 8: Overview of sex control in animal reproduction***	
	A/ Main contents: *(1 lesson)* **Theory:** 8.1. Common problems 8.2. Overview of measures to control sex 8.3. Conclusion **Practice/Experiment:** *(2 lesson)* - Sex segregation and its application in cow intrauterine insemination	K1-K7
	***B/* Self- study contents:** *(4 lesson)* - Methods of controlling sex in cattle	K1-K7

X. Facility and other requirements

- Classrooms, practice rooms: Spacious classrooms, fully equipped for teaching
- Teaching facilities: : Projector, screen, microphone, computers
- Other vehicles: Animals for experimentation

Ha Noi, 202

HEAD OF DEPARTMENT	LECTURER
(Full name and signature)	*(Full name and signature)*
	Ngô Thành Trung
DEAN OF FACULTY	**PRESIDENT**
(Full name and signature)	*(Full name and signature)*

APPENDIX

LIST OF LECTURERS AND ASSISSTANTS FOR THE COURSE

Lecture

Full name: Su Thanh Long	Academic Title, Academic Distinction: Associate Professor Ph.D
Address: Surgery-Reproduction, Veterinary Medicine	Phone: 0904870888
Email: sulongjp@yahoo.com	Website: https://khoathuy.vnua.edu.vn/
Contact with lectures/teaching assisstant: By phone, email	

Lecture

Full name: Nguyen Van Thanh	Academic Title, Academic Distinction: Associate Professor Ph.D
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Lecture

Full name: Nguyen Thi Mai Tho	Academic Title, Academic Distinction: Veterinarians
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Lecture

Full name: Nguyen Cong Toan	Academic Title, Academic Distinction: Master
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Lecture

Full name: Bui Van Dung	Academic Title, Academic Distinction: Veterinarians
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Lecture

Full name: Nguyen Hoai Nam	Academic Title, Academic Distinction: Ph.D
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Lecture

Full name: Do Thi Kim Lanh	Academic Title, Academic Distinction: Ph.D
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Lecture

Full name: Nguyen Duc Truong	Academic Title, Academic Distinction: Master
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Lecture

Full name: Ngo Thanh Trung	Academic Title, Academic Distinction: Master
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SUMMARY TABLE OF COMPATIBILITY BETWEEN COURSE EXPECTED LEARNING OUTCOMES, TEACHING-LEARNING AND ASSESSMENT

COURSE EXPECTED LEARNING OUTCOMES	K1	K2	K3	K4	K5	K6	K7
TEACHING AND LEARNING							
Presentation	x	x	x			x	x
Practical				x	x	x	x
ASSESSMENT							
Rubric 1. Attendance	x	x	x	x	x	x	x
Rubric 2. Practical				x	x	x	x
Rubric 3: Mid-term exam	x	x	x				
Rubric 4: Final exam	x	x	x				

TIMES TO IMPROVE THE MAJOR

- 1st time: July 2018
Number of credits reduced by 1 credit